# JAVIER CHICO VÁZQUEZ

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## EDUCATION

**University of Oxford** — **Oxford, UK** | ***2023 - Present***

- **PhD in Mathematics**. Supervised by Prof. Derek Moulton and Prof. Dominic Vella.

**Massachusetts Institute of Technology** — **Cambridge, MA, USA** | ***2021 - 2022***

- Exchange year at MIT, Department of Mathematics. **5.0 GPA**

**Imperial College London** — **London, UK** | ***2019 - 2023***

- **MSci** in Mathematics — **First Class**, overall **92.65%**
- **Governor's Prize.** Awarded to the best student in Mathematics. (**Ranked first in my year**.)
- **Master Thesis** with Prof. D.T. **Papageorgiou** on ferrofluids.
- **Dean's list** in Y1,Y2,Y3,Y4. 2021 G-RESEARCH academic excellence prize.

## EXPERIENCE

**Mathematical Institute, University of Oxford**, *Doctoral Researcher* — ***September 2023-Present***

- **Fluid Dynamics of the inner ear** The fluid dynamics of human balance are studied to evaluate strategies to mitigate dizziness. The nonlinear PDEs governing the interaction between fluid flow and cupular deflection are reduced asymptotically to a one dimensional dynamical system for the cupular deflection.
- **Pollination in Rafflesia** As part of a collaboration with Chris Thorogood, head of the of the Oxford Botanical Garden, we have developed a model to understand how the biggest flowers in the world attract flies for pollination. The aim of the project is to understand the role of floral shape and size in pollination.
- **Electroreception** We use mathematical methods to understand the electrical sensing capabilities of bees and sharks, as well as their implication on the ecology and locomotion of these animals.
- **Stability of flapping formations** (with C. Mavroyiakoumou) modelling of group flight, with a focus on stability and collision avoidance.
- **Cycling** Developed a mathematical model to predict the optimal breakaway location to win a cycling race.

**Brasenose College, University of Oxford**, *Stipendiary Lecturer* — ***October 2025 - Present***

- **First Year (Prelims):** Geometry, Multivariable Calculus, Dynamics and Computational Mathematics.

**Imperial College London**, *MSc Researcher* — ***October 2022 - May 2023***

- **MSc thesis:** "Ferrofluids falling under gravity".

**Massachusetts Institute of Technology**, *Undergraduate Researcher* — ***January - September 2022***

- UROP with Dr Andrew **Horning** on Spectral Density Estimation and Kernel Polynomial Methods. Developed a new result for the convergence in distribution of random variables transformed by Chebyshev polynomials.
- Summer research with Prof. Lydia **Bourouiba** on the Fluid Dynamics of Disease Transmission. Used network science to study diffusion on porous networks.

**Citigroup Global Markets Limited**, *Summer Analyst & Spring Intern* — **London** | ***Summer 2021***

## PUBLICATIONS AND MANUSCRIPTS

1. Chico-Vázquez, J., Moulton, D. E., & Vella, D. (2025). Uncovering flow and deformation regimes in the coupled fluid–solid vestibular system. *Journal of Fluid Mechanics*, 1022, A40
2. Chico-Vázquez J. and Griffiths I. M. (2025). A mathematical model for optimal breakaways in cycling: balancing energy expenditure and crash risk. *R. Soc. Open Sci.* 12: 250972

3. Chico-Vázquez J. & Mavroyiakoumou C. (2026) Flapping strategies for flying formations. *Under review in Proc. Soc. A.* Preprint

## AWARDS AND ACHIEVEMENTS

- **Governor's Prize.** Awarded to the best MSci student in Mathematics at their final examinations.
- **Excellence in Physics Distinguished Student Prize 2025** American Physical Society ($2500).
- Graduated ranked first in my cohort.
- **UKRI/St John's College Doctoral Scholarship**
- Awarded **best poster** by the jury at Dynamic Days 2024.
- **Turing Grant** Awarded as part of my exchange year at MIT (£3858.57)
- **Imperial College Presidential Scholarship** (Rejected)
- **Madrid Academic Excellence Grant** awarded by the Madrid Education Board (2100 €). (2019)
- **G-RESEARCH prize for academic excellence** in Year 2 (2021).
- **Wilhelm and Else Heraeus Foundation scholarship** to attend Dynamic Days Bremen 2024. (600 €)
- **Landahl Award 2025** from the Society of Mathematical Biology ($700).
- **Oxford 3 Minute Thesis Competition 2024** Audience favourite.
- **Dean's list** in Y1, Y2, Y3 and MSc year awarded for academic excellence.
- Selected for the MIT exchange (one spot in the Department).

## TEACHING EXPERIENCE

**Mathematical Institute, University of Oxford** ***September 2023-Present***

- Tutor for **Perturbation Theory** (2024, 2025), **Mathematical Mechanical Biology** (2025), **Solid Mechanics** (2025).
- College Tutor for **Keble College** in 2025 and 2026 and **Merton College** in Spring 2024.
- Teaching assistant for **Viscous Flow**, **Applied PDEs**, **Mathematical Mechanical Biology**.

## CONFERENCES

1. Society of Mathematical Biology Annual Meeting 2025.
2. APS Global Physics Meeting 2025, Anaheim.
3. Dynamic Days 2024, Bremen. Awarded Best poster.
4. SIAM Oxford 3 Minute Thesis 2024, Oxford. Audience favourite.

## PROJECTS

- **Master Thesis** Ferrofluids on cylindrical domains with Prof. Papageorgiou.
- **Year 2 Research:** *The 1:1:2 resonance and the stepwise precession of the swing plane*, supervised by Prof. D. Holm. Developed a semi-empirical formula for the precession angle and studied the effects of including stochasticity into the classical problem of the elastic spherical pendulum.
- **Mathematics of Pelotons** As part of MIT's 18.355 Fluid Dynamics by Prof. Bush I research the aerodynamics of groups of cyclists, and studied their optimal shape, and the optimal position within the peloton.
- **Pattern formation in growing domains** under the guidance of Dr. Ousmane Kodio at MIT.
- **Weakly nonlinear analysis of SIS epidemic system** presented as the research project for 18.377 at MIT, under the supervision of Prof. Akylas.
- **ICDSS Insurance Pricing Competition 2020** Second prize in the nonlinear model category (2020). I used generalised linear models to build a predictive model from data from 80000 real drivers.